

Math Class 9th Federal Board Chapter

02 Logarithms Revision and Full Test

[SLO: M-09-A-06]: Describe logarithm of a number. X	Formative	K	Question(s) will not be asked in the annual examination;
✓ [SLO: M-09-A-07]: Differentiate between common and natural logarithm.	Summative	K	Question(s) will be asked in the annual examination
✓ [SLO: M-09-A-08]: Apply laws of logarithm to real life situations such as growth and decay, loudness of sound.	Summative	A	Question(s) will be asked in the annual examination
[SLO: M-09-A-09]: Apply concepts of rational numbers to real word problems (such as inventory (stock taking), temperature, banking, measures of gain and loss, sources of income and expenditure).	Summative	A	Question(s) will be asked in the annual examination

Q.1 Choose the correct option. Each part carries 1 mark.

1. If $\log(x + 3) = \log(15x - 4)$, then x is:

- (A) 0.5
(C) 14

- (B) 7
(D) 2

$$x+3=15x-4$$

2. $\log_7(7^{-3}) + \log_2(4^3)$ is:

- (A) 3
(C) 0

- (B) -3
(D) ± 3

$$14x = 7$$

$$x = \frac{7}{14} = \frac{1}{2}$$

3. $\log_b(M + N)$ is:

- (A) $\log_b(MN)$
(C) Both (A) and (B)

- (B) $\log_b M + \log_b N$
(D) None of these

4. $\log_b(g^h)$ is:

- (A) $g \log_b h$
(C) $h \log_b g$

- (B) $\log_b(gh)$
(D) $h \log_g b$

$\log_{10}$
 $\ln$ or
 $\log_e$

$$\log_a x^m = m \log_a x$$

$$= -3 \log_7 7 + \log_2 4^3$$

$$= -3 + 6 = 3$$

$$\log_a a = 1$$

$$\ln e = 1$$

$$\log_e e = 1$$

$$\log_a 1 = 0$$

$$\log_5 1 = 0, \quad \ln 1 = 0$$

3

$$\log_a(xy) = \log_a x + \log_a y$$

$$\log_a(x+y) \quad \times$$

5. $\log_b M - \log_b N$ is:

(A) $\frac{\log_b M}{\log_b N}$

(C) $\log_N M$

(B) $\log_b \left(\frac{M}{N} \right)$

(D) $\frac{\log_b N}{\log_b M}$

5. $\log_b M - \log_b N$ is:

- (A) $\frac{\log_b M}{\log_b N}$
(C) $\log_N M$

(B) $\log_b \left(\frac{M}{N} \right)$ ✓
(D) $\frac{\log_b N}{\log_b M}$

6. $\log_{\sqrt{10}}(100^2)$ is:

- (A) 2
(C) 4

(B) 1
(D) 8 ✓

7. $\log 18$ is:

- (A) $3 \log 2 + \log 3$
(C) $3 \log 3 + 2 \log 2$

(B) $\log 2 + 2 \log 3$
(D) $2 \log 3 + 3 \log 2$

8. If $\log_{\sqrt{x}} 25 = 4$, then x is:

- (A) +5 ✓
(C) ± 5

- (B) -5
(D) Impossible

$$\begin{array}{r} 2 \overline{) 18} \\ \underline{4} \\ 3 \\ \underline{6} \\ 3 \\ \underline{3} \\ 0 \end{array}$$

$$18 = 2 \times 3^2$$

$$\log 18 = \log (2 \times 3^2)$$

$$= \log 2 + \log 3^2$$

$$= \log 2 + 2 \log 3$$

Non-recurring + Non-Term

9. The base of the natural logarithm is:

- (A) Rational number
(C) Irrational number

- (B) Integer
(D) 10

10. $\log_{\sqrt{b}}(10^4) \div \log_{\sqrt{b}}(10)$ is:

- (A) $\log_{\sqrt{b}} \left(\frac{10^4}{10} \right)$
(C) 4

- (B) $\log_{\sqrt{b}}(10^4) - \log_{\sqrt{b}}(10)$
(D) $\log_{\sqrt{b}}(10^4 - 10)$

(A) $\log_{\sqrt{b}} \left(\frac{10^4}{10} \right)$

(C) 4

(B) $\log_{\sqrt{b}}(10^4) - \log_{\sqrt{b}}(10)$

(D) $\log_{\sqrt{b}}(10^4 - 10)$

$\log_e x = \ln x$

$\log_{\sqrt{b}} 10^4 = \frac{4 \log 10}{\log \sqrt{b}} = \frac{4 \log 10}{\frac{1}{2} \log b} = 8 \frac{\log 10}{\log b}$

$\log_{\sqrt{b}} 10 = \frac{\log 10}{\log \sqrt{b}} = \frac{\log 10}{\frac{1}{2} \log b} = 2 \frac{\log 10}{\log b}$

$\log_{\sqrt{b}}(10^4 - 10)$

11. $5 \log 2 - 2 \log 5$ is:

(A) $\frac{(\log 2)^5}{(\log 5)^2}$

(C) $\log \left(\frac{2^5}{5^2} \right)$

(B) $\frac{\log 2^5}{\log 5^2}$

(D) $\frac{5}{2} \log \left(\frac{2}{5} \right)$

12. $\log_y x$ will be equal to:

(A) $\frac{\log_z x}{\log_y z}$

(C) $\frac{\log_z x}{\log_z y}$

(B) $\frac{\log_x z}{\log_y z}$

(D) $\frac{\log_z y}{\log_z x}$

$\log 2^5 - \log 5^2$

$= \log \left(\frac{2^5}{5^2} \right)$

12. $\log_y x$ will be equal to:

(A) $\frac{\log_z x}{\log_y z}$

(C) $\frac{\log_z x}{\log_z y}$

(B) $\frac{\log_x z}{\log_y z}$

(D) $\frac{\log_z y}{\log_z x}$

$$\textcircled{C} \frac{\log_y z}{\log_z x} \longleftrightarrow \textcircled{D} \frac{\log_y z}{\log_z y}$$

change of base

$$\log_y x = \frac{\log_z x}{\log_z y}$$

$$\log_5 15 = \frac{\log_{10} 15}{\log_{10} 5}$$

13. $\log_b a \times \log_c b$ can be written as:

α (A) $\log_a c$

α (C) $\log_a b$

α (B) $\log_c a$

(D) $\log_b c$

$$\log_b a \times \log_c b = \log_c a$$

$$\log_b a = \frac{\log_e a}{\log_e b}$$

14. The relation $y = \log_z x$ implies:

(A) $x^y = z$ ~~z~~ ~~z~~

(C) $x^z = y$

(B) $z^y = x$

(D) $y^z = x$

15. If $a^x = n$, then:

(A) $a = \log_x n$

(C) $x = \log_a n$

(B) $x = \log_n a$

(D) $a = \log_n x$

14/1 $y = \log_z x$
 $z = \log_x x$
 $x = z^y$

15. If $a^x = n$, then:

(A) $a = \log_x n$

(C) $x = \log_a n$

(B) $x = \log_n a$

(D) $a = \log_n x$

$\log_a a^x = \log_a n$
 $x \log_a a = \log_a n$

Imp $\checkmark$

(i) A radioactive substance decays according to

$$M = 200 \times 10^{-0.05t},$$

where M is the mass in grams and t is the time in years.

(a) Use logarithms to express t in terms of M .

(b) Find the time t when the mass M reduces to 100 grams.

n Given

$M = 200 \times 10^{-0.05t}$

$-0.05t$

a) Given $M = 200 \times 10^{-0.05t}$

$$\Rightarrow \frac{M}{200} = 10^{-0.05t}$$

$$\Rightarrow \log\left(\frac{M}{200}\right) = \log 10^{-0.05t}$$

$$\Rightarrow -0.05t \log 10 = \log\left(\frac{M}{200}\right)$$

$$-0.05t = \log\left(\frac{M}{200}\right)$$

$$t = -\frac{1}{0.05} \times \log\left(\frac{M}{200}\right)$$

$$t = -20 \log\left(\frac{M}{200}\right)$$

$$t = 20 \log\left(\frac{M}{200}\right)^{-1}$$

$$\boxed{1 - \dots \log(200)} \rightarrow \textcircled{1}$$

$$\boxed{t = 20 \log\left(\frac{200}{M}\right)} \rightarrow \textcircled{1}$$

b) Given $M = 100g$, $t = ?$

$$t = 20 \log\left(\frac{200}{100}\right)$$

Thus, $\boxed{t \approx 6.02 \text{ Years.}}$

- (ii) Bani Gala had a population of 10,000 people in the year 2015. The population has been growing exponentially at a rate of 2.5% per year. Using the exponential growth formula

$$\underline{r} \quad P(t) = P_0 e^{rt}, \rightarrow \textcircled{1}$$

apply laws of logarithms to determine the year when the population reaches 25,000.

Sol: Given, $P_0 = 10,000$

$$r = 2.5\% = \frac{2.5}{100} = 0.025$$

$$P(t) = 25,000$$

To find: $t = ?$

Eqn. (1) becomes.

$$25,000 = 10,000 e^{0.025t}$$

$$\Rightarrow e^{0.025t} = \frac{25,000}{10,000}$$

$$\Rightarrow e^{0.025t} = 2.5$$

Apply Natural log (ln) on
b.s.

$$\ln e^{0.025t} = \ln 2.5$$

$$0.025t \times \ln e = \ln 2.5$$

$$t = \frac{\ln 2.5}{0.025}$$

$$t = \underline{\underline{20.65 \text{ years}}}$$

$$P = 10,000$$

In 2015 $\rightarrow P_0 = 10,000$ appx
 NW Reg. Year = $2015 + 36 = 2051$ ✓
 or Reg. Year = 2052 appx ✓

(iv) An earthquake that occurred in Pakistan in 2005 measured 7.6 on the Richter scale, while an earthquake in China measured 8.2. How many times stronger was the earthquake in China than the earthquake in Pakistan? Use

$$M = \log \left(\frac{I}{I_0} \right) \rightarrow \textcircled{1}$$

Let for Pakistan Earthquake.
 Intensity $\leftarrow I = I_1$, $M = 7.6$ ✓

$$\textcircled{1} \Rightarrow 7.6 = \log \left(\frac{I_1}{I_0} \right) \rightarrow \textcircled{2}$$

For China's Earthquake

$$I = I_2, M = 8.2$$

$$\textcircled{1} \Rightarrow 8.2 = \log \left(\frac{I_2}{I_0} \right) \rightarrow \textcircled{3}$$

Subtract $\textcircled{2}$ from $\textcircled{3}$, we get

$$\log\left(\frac{I_2}{I_0}\right) - \log\left(\frac{I_1}{I_0}\right) = 8.2 - 7.6$$

$$\log I_2 - \log I_0 - (\log I_1 - \log I_0) = 0.6$$

$$\log I_2 - \cancel{\log I_0} - \log I_1 + \cancel{\log I_0} = 0.6$$

$$\log I_2 - \log I_1 = 0.6$$

$$\Rightarrow \log\left(\frac{I_2}{I_1}\right) = 0.6$$

Take Antilog on both sides

$$10^{\log\left(\frac{I_2}{I_1}\right)} = 10^{0.6}$$

$$\boxed{\frac{I_2}{I_1} = 3.98 \approx 4}$$

Here Richter's Earthquake

Hew. China's Earthquake
is approx. 4 times stronger
than Pakistan's Earthquake

(vii) Find the number of digits in 5^{50} .

Sol.

Number of digits in 5^{50}

$$= \overset{\text{char.}}{\log} 5^{50} + 1$$

$$= 50 \times \log 5 + 1$$

$$= 50 \times 0.699 + 1$$

$$= \underline{\underline{34.9485}} + 1$$

$$= 34 + 1 \quad \left(\begin{array}{l} \therefore \text{Characteristic} \\ \therefore \text{Ans} = 34 \end{array} \right)$$

$$= 35$$

50 ..

$= \Rightarrow$
Thus 1 total digits in $5^{50} = 35$

(ix) Given

$A = A_0 e^{-kd} \rightarrow \textcircled{1}$

If $k = 2$, what should be the value of d to make $A = \frac{A_0}{2}$?

Sol. put given values in (1), we get

$$\frac{A_0}{2} = A_0 e^{-2d}$$

$$\Rightarrow \frac{1}{2} = e^{-2d}$$

$$\Rightarrow \ln\left(\frac{1}{2}\right) = \ln e^{-2d}$$

$$\ln 1 - \ln 2 = -2d \times \ln e$$

$$- \ln 2 = -2d$$

$$\Rightarrow 2d = \ln 2$$

$$\Rightarrow 2d = \ln 2$$

$$\boxed{d = \frac{\ln 2}{2}}$$

2nd last

(viii) Use laws of logarithms to evaluate:

$$\frac{\sqrt[3]{8.59} \times (55.6)^2}{2.51 \times \sqrt{2.12}}$$

Sol. Let $x = \frac{\sqrt[3]{8.59} \times (55.6)^2}{2.51 \times \sqrt{2.12}}$

$$\Rightarrow \log x = \log \left(\frac{\downarrow}{\downarrow} \right)$$

$$= \log(\sqrt[3]{8.59} \times (55.6)^2) - \log(2.51 \sqrt{2.12})$$

$$= \log \sqrt[3]{8.59} + \log (55.6)^2 - [\log 2.51 + \log \sqrt{2.12}]$$

$$= \log (8.59)^{\frac{1}{3}} + 2 \log (55.6) - \log 2.51 - \frac{1}{2} \log 2.12$$

$$= \log(8.59)^2 + 2\log(55.6) - \log 2.1 - \frac{1}{2}\log(2.12)$$

$$= \frac{1}{3}\log(8.59) + 2\log(55.6) - \log 2.1 - \frac{1}{2}\log(2.12)$$

$$\log x = 3.2386$$

Take Antilog on B.S.

$$x = 10^{3.2386}$$

$$x = 1732.36$$

(x) Evaluate:

Let

$$x = \sqrt[3]{\frac{0.07921 \times (18.99)^2}{(5.79)^4 \times 0.9474}}$$

$$\log x = \log \left(\frac{0.07921 \times (18.99)^2}{(5.79)^4 \times 0.9474} \right)^{\frac{1}{3}}$$

$$\log x = \frac{1}{3} \log \left(\frac{\quad}{\quad} \right)$$

$$= \frac{1}{3} \left[\log(0.07921 \times (18.99)^2) - \log((15.79)^4 \times 0.9579) \right]$$

$$= \frac{1}{3} \left[\log(0.07921) + 2\log(18.99) - \{4\log 15.79 + \log 0.9579\} \right]$$

$$= \frac{1}{3} \left[\log(0.07921) + 2\log(18.99) - 4\log 15.79 - \log 0.9579 \right]$$

$$= \frac{1}{3} \left[\begin{array}{l} \underline{\underline{-2.8988}} + 2.5570 - 3.05071 - \begin{array}{l} \underline{\underline{1.9765}} \\ \uparrow \downarrow \\ -ve \quad +ve \end{array} \end{array} \right]$$

$$\log x = \frac{1}{3} \left[-2 + 0.8988 + 2.5570 - 3.05071 + 1 - 0.9765 \right]$$

$$\Rightarrow \log x = \frac{1}{3} (-1.57141)$$

Take Antilog on BS. —

Take $\ln(1.5711)$

$$\begin{array}{r} 0.5238 \\ -1.5711 \\ \hline \end{array}$$

$$x = 10$$

$$\boxed{x = 0.02683}$$

$$\boxed{x = 0.2994} \quad V$$

[Signature]